1·2 Gaussian Elimination

Example 1:

Row Echelon and Reduce Row Echelon Form.

$$\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix} \rightarrow \text{Row Echelon.}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↳ Reduce Row Echelon

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ Row Echelon

$$\begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

↳ Row Echelon.

$$\begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -7 \end{bmatrix}$$

↳ RREF

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ RREF.

Example: 3

Unique Solution. $\leftarrow$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 5 \end{bmatrix} \quad \begin{aligned} x_1 &= 3 \\ x_2 &= -1 \\ x_3 &= 0 \\ x_4 &= 5 \end{aligned}$$

Example : 4

Linear system in Unknowns

a-

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow 0 \neq 1$$

↳ No Solution (Inconsistent)

b-

Infinite Solution ←

$$\begin{bmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -4 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$x + 0y + 3z = -1 \rightarrow (1)$

$0x + 1y - 4z = 2 \rightarrow (2)$

$\rightarrow 0 = 0$

Leading variables $= x, y$

Free variables $= z$.

Let $\boxed{z = t}$

From eq (1)

$x + 3z = -1$

$x + 3t = -1$

$\boxed{x = -1 - 3t}$

From eq (2)

$y - 4z = 2$

$\boxed{y = 2 + 4t}$

$\boxed{t \in \mathbb{R}}$

c-

Infinite Solution. ←

$$\begin{bmatrix} 1 & -5 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow x - 5y + z = 4 \rightarrow (1)$$

Leading variables $= x$

Free variables $= y, z$.

Let $\boxed{y = s}$ $\boxed{z = t}$

From eq (1)

$x - 5y + z = 4$

$\boxed{x = 4 + 5s - t}$

$\boxed{s, t \in \mathbb{R}}$

Exercise Set - 1.2

Question: 1

Determine whether matrix is in REF, RREF, both or niether.

a-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↳ Both

b-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ Both.

c-

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ Both.

d-

$$\begin{bmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 2 & 4 \end{bmatrix}$$

↳ Both

e-

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

↳ Both

f-

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

↳ Both.

g-

$$\begin{bmatrix} 1 & -7 & 5 & 5 \\ 0 & 1 & 3 & 2 \end{bmatrix}$$

↳ Row Echelon Form.

Question: 2

Determine whether matrix is in REF, RREF, both or niether.

a-

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ REF

b-

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix}$$

↳ Niether

c-

$$\begin{bmatrix} 1 & 3 & 4 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ REF

e-

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

↳Niether

f-

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 0 & 7 & 1 & 3 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

↳Niether

g-

$$\begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

↳Both.

d-

$$\begin{bmatrix} 1 & 5 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

↳ REF

Question: 3

Suppose-Augmented matrix has been reduced by row operations to given row echelon form. Solve System.

a-

$$\begin{bmatrix} 1 & -3 & 4 & 7 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}$$

$$x - 3y + 4z = 7 \rightarrow (1)$$
$$y + 2z = 2 \rightarrow (2)$$
$$\boxed{z = 5} \rightarrow (3)$$

put eq (3) in eq (2)

$$y + 2(5) = 2$$
$$\boxed{y = -8}$$

put $y = -8$ and $z = 5$ in (1)

$$x - 3(-8) + 4(5) = 7$$
$$x + 24 + 20 = 7$$
$$\boxed{x = -37}$$

Unique Solution.

b-

$$\begin{bmatrix} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$$

Leading Variables = x, y, w
Free Variable = z.

$$w \quad + 8y - 5z = 6 \rightarrow (1)$$
$$x + 4y - 9z = 3 \rightarrow (2)$$
$$y + z = 2 \rightarrow (3)$$

Let

$z = t$

From eq (3)

$y + z = 2$

$\boxed{y = 2 - t}$

From eq (2)

$x + 4y - 9z = 3$

$x = 3 - 4(2-t) + 9t$

$x = 3 - 8 + 4t + 9t$

$\boxed{x = 13t - 5}$

For eq (1)

$w + 8y - 5z = 6$

$w = 6 - 8(2-t) + 5t$

$w = 6 - 16 + 8t + 5t$

$\boxed{w = 13t - 10}$

Infinite Solutions.

$t \in \mathbb{R}$.

C-

$$\begin{bmatrix} 1 & 7 & -2 & 0 & -8 & -3 \\ 0 & 0 & 1 & 1 & 6 & 5 \\ 0 & 0 & 0 & 1 & 3 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Leading variables $= x_1, x_3, x_4$

Free variables $= x_2, x_5$

$$x_1 + 7x_2 - 2x_3 \quad - 8x_5 = -3 \rightarrow (1)$$
$$x_3 + x_4 + 6x_5 = 5 \rightarrow (2)$$
$$x_4 + 3x_5 = 9 \rightarrow (3)$$

Let

$\boxed{x_2 = s}, \boxed{x_5 = t}$

From eq 3

$x_4 + 3t = 9$

$\boxed{x_4 = 9 - 3t}$

From eq 2

$x_3 + (9 - 3t) + 6t = 5$

$x_3 = 5 - 9 - 6t + 3t$

$\boxed{x_3 = -4 - 3t}$

From eq (1)

$$x_1 + 7x_2 - 2x_3 - 8x_5 = -3$$

$$x_1 + 7s - 2(-4-3t) - 8t = -3$$

$$x_1 + 7s + 8 + 6t - 8t = -3$$

$$\boxed{x_1 = -11 - 7s + 2t}$$

Infinite Solution

$s, t \in \mathbb{R}$.

d-

$$\begin{bmatrix} 1 & -3 & 7 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$\hookrightarrow$ Last Row $0 \neq 1$ (contradiction)

No Solution

Question: 04

Suppose Augmented matrix has been reduce by row operation to given row echelon form. Solve System

a-

$$\begin{bmatrix} 1 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 7 \end{bmatrix}$$

Leading variables $= x_1, x_2, x_3$

Free Variable = none.

Unique Solution.

$\boxed{x_1 = -3}$, $\boxed{x_2 = 0}$

$\boxed{x_3 = 7}$

b-

$$\begin{bmatrix} 1 & 0 & 0 & -7 & 8 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 1 & -5 \end{bmatrix}$$

Leading variables $= x_1, x_2, x_3$
Free variable $= x_4$

$$\begin{aligned} x_1 \qquad -7x_4 &= 8 \rightarrow (1) \\ x_2 \quad +3x_4 &= 2 \rightarrow (2) \\ x_3 + x_4 &= -5 \rightarrow (3) \end{aligned}$$

Let

$x_4 = t$

From eq. (3)

$x_3 + t = -5$

$x_3 = -5 - t$

From eq. (1)

$x_1 - 7t = 8$

$x_1 = 8 + 7t$

From eq. (2)

$x_2 + 3x_4 = 2$

$x_2 + 3t = 2$

$x_2 = 2 - 3t$

Infinite solutions.

$t \in \mathbb{R}$.

c-

$$\begin{bmatrix} 1 & -6 & 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Leading variables $= x_1, x_3, x_4$
Free variables $= x_2, x_5$

$$\begin{aligned} x_1 - 6x_2 \qquad +3x_5 &= -2 \rightarrow (1) \\ x_3 \quad +4x_5 &= 7 \rightarrow (2) \\ x_4 + 5x_5 &= 8 \end{aligned}$$

Let

$x_2 = s$, $x_5 = t$

From eq (3)

$x_4 + 5x_5 = 8$

$x_4 = 8 - 5t$

From eq (2)

$x_3 + 4x_5 = 7$

$x_3 = 7 - 4t$

From eq (1)

$x_1 - 6x_2 + 3x_5 = 2$

$x_1 - 6s + 3t = 2$

$x_1 = 2 + 6s - 3t$

Infinite solutions

$s, t \in \mathbb{R}$

d-

$$\begin{bmatrix} 1 & -3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

↳ Last row $0 \neq 1$ (contradiction)

No solution.

Question: 5-8

Solve the linear system by Gaussian Elimination method.

5-

$$x_1 + x_2 + 2x_3 = 8$$
$$-x_1 - 2x_2 + 3x_3 = 1$$
$$3x_1 - 7x_2 + 4x_3 = 10$$

Taking Augmented Matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right] \begin{array}{l} \\ R_2 + R_1 \\ R_3 - 3R_1 \end{array}$$

$$-R_2\begin{bmatrix} 1 & 1 & 2 & | & 8 \\ 0 & 1 & -5 & | & -9 \\ 0 & -10 & -2 & | & -14 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 2 & | & 8 \\ 0 & 1 & -5 & | & -9 \\ 0 & 0 & -52 & | & -104 \end{bmatrix} R_3+10R_2$$

$$\begin{bmatrix} 1 & 1 & 2 & | & 8 \\ 0 & 1 & -5 & | & -9 \\ 0 & 0 & 1 & | & 2 \end{bmatrix} -\frac{1}{52}R_3$$

→ Row Echelon Form

$$x_1+x_2+2x_3 = 8 \rightarrow (1)$$
$$x_2 - 5x_3 = -9 \rightarrow (2)$$
$$\boxed{x_3 = 2} \rightarrow (3)$$

put $x_3=2$ in eqn (2)

$$x_2 - 5x_3 = -9$$
$$x_2 - 5(2) = -9$$
$$x_2 = -9+10$$
$$\boxed{x_2 = 1}$$

put $x_2=1$ and $x_3=2$ in eqn (1)

$$x_1+x_2+2x_3 = 8$$
$$x_1+1+4 = 8$$
$$\boxed{x_1 = 3}$$

Unique solution

6-

$$2x_1+2x_2+2x_3 = 0$$
$$-2x_1+5x_2+2x_3 = 1$$
$$8x_1+x_2+4x_3 = -1$$

Taking Augmented Matrix

$$\begin{bmatrix} 2 & 2 & 2 & | & 0 \\ -2 & 5 & 2 & | & 1 \\ 8 & 1 & 4 & | & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & | & 0 \\ -2 & 5 & 2 & | & 1 \\ 8 & 1 & 4 & | & -1 \end{bmatrix} \frac{1}{2}R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right] \begin{array}{l} R_2+2R_1 \\ R_3-8R_1 \end{array} \qquad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 4/7 & 1/7 \\ 0 & 7 & -4 & -1 \end{array}\right] \frac{1}{7}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 4/7 & 1/7 \\ 0 & 0 & 0 & 0 \end{array}\right] R_3+7R_2$$

→ Last Row becomes equal to zero so system has infinite solutions.

Leading variables = x_1, x_2
Free variables = x_3

$$x_1+x_2+x_3=0 \rightarrow (1)$$
$$x_2+\frac{4}{7}x_3=\frac{1}{7} \rightarrow (2)$$

Let

$\boxed{x_3=t}$

From eq (1)

$$x_2+\frac{4}{7}x_3=\frac{1}{7}$$
$$x_2=\frac{1}{7}-\frac{4}{7}x_3$$
$$\boxed{x_2=\frac{1}{7}-\frac{4}{7}t}$$

From eq (2)

$$x_1+\left(\frac{1}{7}-\frac{4}{7}t\right)+t=0$$
$$x_1+\frac{1}{7}-\frac{4}{7}t+t=0$$
$$x_1=\frac{-1}{7}+\frac{4}{7}t-t$$
$$x_1=\frac{-1+4t-7t}{7}$$
$$\boxed{x_1=\frac{1}{7}-\frac{3}{7}t}$$

$t \in \mathbb{R}$.

7 -

$$x-y+2z-w=-1$$
$$2x+y-2z-2w=-2$$
$$-x+2y-4z+w=1$$
$$3x \qquad -3w=-3$$

Taking Augmented Matrix

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \\ 3 & 0 & 0 & -3 & 3 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 3 & -6 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 3 & -6 & 0 & 0 \end{array}\right] \begin{array}{l} R_2-2R_1 \\ R_3+R_1 \\ R_4-3R_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 3 & -6 & 0 & 0 \end{array}\right] \frac{1}{3}R_2 \quad \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} R_3-R_2 \\ R_4-3R_2 \end{array}$$

Last 2 rows becomes equal to zero so system has infinite solutions.

Leading Variables = x, y
Free variables = z, w

$$x - y + 2z - w = -1 \rightarrow (1)$$
$$y - 2z = 0 \rightarrow (2)$$

Let
$\boxed{z = s}$
$\boxed{w = t}$

From eq (2)
$y - 2z = 0$
$y - 2s = 0$
$\boxed{y = 2s}$

From eq (1)
$x = -1 + y - 2z + w$
$x = -1 + 2s - 2s + t$
$\boxed{x = -1 + t}$

$s, t \in \mathbb{R}$.

$$-2b + 3c = 1$$
$$3a + 6b - 3c = -2$$
$$6a + 6b + 3c = 5$$

Taking Augmented Matrix

$$\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] \frac{1}{3}R_1 \quad \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & -6 & 9 & 9 \end{array}\right] R_3 - 6R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & -6 & 9 & 9 \end{array}\right] -\frac{1}{2}R_1 \quad \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 6 \end{array}\right]$$

↳ Last row $0 \neq 6$

No Solution.

Question: Q-12-

Solve the linear system by Gauss-Jordan elimination.

a-

$$x_1 + x_2 + 2x_3 = 8$$
$$-x_1 - 2x_2 + 3x_3 = 1$$
$$3x_1 - 7x_2 + 4x_3 = 10$$

Taking Augmented Matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right] \begin{array}{l} R_2+R_1 \\ R_3-3R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right] -R_2 \quad \left[\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right] \begin{array}{l} R_1-R_2 \\ R_3+10R_1 \end{array}$$

$$\left[\begin{array}{cccc} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right] -\tfrac{1}{52}R_3 \quad \left[\begin{array}{cccc} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \begin{array}{l} R_2+5R_3 \\ R_1-7R_3 \end{array}$$

↳ Reduce Echelon Form.

$\boxed{x_1 = 3}$, $\boxed{x_2 = 1}$

$\boxed{x_3 = 2}$

10 -

$$2x_1+2x_2+2x_3 = 0$$
$$-2x_1+5x_2+2x_3 = 1$$
$$8x_1+x_2+4x_3 = -1$$

Taking Augmented Matrix:

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \tfrac{1}{2}R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right] \begin{array}{l} R_2+2R_1 \\ R_3-8R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 4/7 & 1/7 \\ 0 & -7 & -4 & -1 \end{array}\right] \tfrac{1}{7}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 3/7 & -1/7 \\ 0 & 1 & 4/7 & 1/7 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} R_1 - R_2 \\ R_3 + 7R_2 \\ \end{array}$$

Leading variables $= x_1, x_2$
Free variable $= x_3$

↳ Last row become 0

$$x_1 + \tfrac{3}{7}x_3 = -\tfrac{1}{7} \rightarrow (1)$$
$$x_2 + \tfrac{4}{7}x_3 = \tfrac{1}{7} \rightarrow (2)$$

Let

$\boxed{x_3 = t}$

From eq (2)

$x_2 + \tfrac{4}{7}t = \tfrac{1}{7}$

$\boxed{x_2 = \tfrac{1}{7} - \tfrac{4}{7}t}$

From eq (1)

$x_1 + \tfrac{3}{7}t = -\tfrac{1}{7}$

$\boxed{x_1 = -\tfrac{1}{7} - \tfrac{3}{7}t}$

$t \in \mathbb{R}$.

11-

$$x - y + 2z - w = -1$$
$$2x + y - 2z - 2w = -2$$
$$-x + 2y - 4z + w = 1$$
$$3x \qquad\quad - 3w = -3$$

Taking Augmented Matrix.

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \\ 3 & 0 & 0 & -3 & -3 \end{array}\right] \rightarrow \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 3 & -6 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 3 & -6 & 0 & 0 \end{array}\right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 + R_1 \\ R_4 - 3R_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 3 & -6 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{3}R_2} \left[\begin{array}{ccccc} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} R_1+R_2 \\ R_3-R_2 \\ R_4-3R_2 \end{array}$$

Leading variables = x, y
Free variables = z, w

$$x \quad -w = -1 \rightarrow (1)$$
$$y - 2z \quad = 0 \rightarrow (2)$$

Let

$\boxed{w=t}$, $\boxed{z=s}$

From eq (2)

$y = 2z$

$\boxed{y = 2s}$

From eq (1)

$x = -1 + w$

$x = -1 + t$

$\boxed{x = t-1}$

$s, t \in \mathbb{R}$

12-

$$-2b + 3c = 1$$
$$3a + 6b - 3c = -2$$
$$6a + 6b + 3c = 5$$

Taking Augmented Matrix

$$\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] \frac{1}{3}R_1 \quad \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & -6 & 9 & 9 \end{array}\right] R_3-6R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2/3 \\ 0 & 1 & -3/2 & -1/2 \\ 0 & -6 & 9 & 9 \end{array}\right] \xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{2} & 1/3 \\ 0 & 1 & -3/2 & -1/2 \\ 0 & 0 & 0 & 6 \end{array}\right] \begin{array}{l} R_1-2R_2 \\ R_3+6R_2 \end{array}$$

↳ Last row $0 \neq 6$

So No Solution.

Question: 19 & 21

Homogenuous

1. Always Consistent (coz 0 is it solution in every case)
2. $X = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (Trivial Solution)
3. If unique then that will always be zero other wise infinite.

19-

$$\begin{aligned} 2x + 2y + 4z &= 0 \\ w \qquad - y - 3z &= 0 \\ 2w + 3x + y + z &= 0 \\ -2w + x + 3y - 2z &= 0 \end{aligned}$$

Taking augmented matrix

$$\left[\begin{array}{cccc|c} 0 & 2 & 2 & 4 & 0 \\ 1 & 0 & -1 & -3 & 0 \\ 2 & 3 & 1 & 1 & 0 \\ -2 & 1 & 3 & -2 & 0 \end{array}\right] \quad \left[\begin{array}{cccc|c} 1 & 0 & -1 & -3 & 0 \\ 0 & 2 & 2 & 4 & 0 \\ 2 & 3 & 1 & 1 & 0 \\ -2 & 1 & 3 & -2 & 0 \end{array}\right] R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -3 & 0 \\ 0 & 2 & 2 & 4 & 0 \\ 0 & 3 & 3 & 7 & 0 \\ 0 & 1 & 1 & -8 & 0 \end{array}\right] \begin{array}{l} R_3-2R_1 \\ R_4+2R_1 \end{array} \quad \left[\begin{array}{cccc|c} 1 & 0 & -1 & -3 & 0 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 3 & 3 & 7 & 0 \\ 0 & 1 & 1 & -8 & 0 \end{array}\right] \frac{1}{2}R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & -3 & 0 \\ 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -10 & 0 \end{array}\right] \begin{array}{l} R_3-3R_2 \\ R_4-R_2 \end{array} \quad \left[\begin{array}{cccc|c} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{l} R_4+10R_3 \\ R_2-2R_3 \\ R_1+3R_3 \end{array}$$

Remark:

1. If no of unknows > no of eq.s then no unique Solution (either no sol or infinite sol)

2. In case of Homogenuous (If no of unknowns > no of eq.s) then always inifinte sol.

Leading variables = w, x, z
Free variables = y

$$w - y = 0 \rightarrow (1)$$
$$x + y = 0 \rightarrow (2)$$
$$z = 0 \rightarrow (3)$$

Let

$y = t$

From eq (2)

$x + y = 0$

$\boxed{x = -t}$

From eq (1)

$w - y = 0$

$w - t = 0$

$\boxed{w = t}$

$\boxed{z = 0}$

$t \in \mathbb{R}$

21-

$$2I_1 - I_2 + 3I_3 + 4I_4 = 9$$
$$I_1 - 2I_3 + 7I_4 = 11$$
$$3I_1 - 3I_2 + I_3 + 5I_4 = 8$$
$$2I_1 + I_2 + 4I_3 + 4I_4 = 10$$

Taking Augmented Matrix

$$\left[\begin{array}{cccc|c} 2 & -1 & 3 & 4 & 9 \\ 1 & 0 & -2 & 7 & 11 \\ 3 & -3 & 1 & 5 & 8 \\ 2 & 1 & 4 & 4 & 10 \end{array}\right]$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 7 & 11 \\ 2 & -1 & 3 & 4 & 9 \\ 3 & -3 & 1 & 5 & 8 \\ 2 & 1 & 4 & 4 & 10 \end{array}\right] R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 7 & 11 \\ 0 & -1 & 7 & -10 & -13 \\ 0 & -3 & 7 & -16 & -25 \\ 0 & 1 & 8 & -10 & -12 \end{array}\right] \begin{array}{l} R_2-2R_1 \\ R_3-3R_1 \\ R_4-2R_1 \end{array}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 7 & 11 \\ 0 & 1 & -7 & 10 & 13 \\ 0 & -3 & 7 & -16 & -25 \\ 0 & 1 & 8 & -10 & -12 \end{array}\right] -R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & -2 & 7 & 11 \\ 0 & 1 & -7 & 10 & 13 \\ 0 & 0 & -14 & 14 & 14 \\ 0 & 0 & 15 & -20 & -25 \end{array}\right] \begin{array}{l} \\ R_3+3R_2 \\ R_4-R_2 \\ \\ \end{array} \left[\begin{array}{cccc|c} 1 & 0 & -2 & 7 & 11 \\ 0 & 1 & -7 & 10 & 13 \\ 0 & 0 & 1 & -1 & 14 \\ 0 & 0 & 15 & -20 & -25 \end{array}\right] -\frac{1}{14}R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 5 & 9 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & -5 & -10 \end{array}\right] \begin{array}{l} \\ R_1+2R_3 \\ R_2+7R_3 \\ R_4-15R_3 \end{array} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 5 & 9 \\ 0 & 1 & 0 & 3 & 6 \\ 0 & 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right] -\frac{1}{5}R_4$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array}\right] \begin{array}{l} R_1-5R_4 \\ R_2-3R_4 \\ R_3+R_4 \\ \\ \end{array}$$

$\boxed{x_1 = -1}$ $\boxed{x_2 = 0}$

$\boxed{x_3 = 1}$ $\boxed{x_4 = 2}$

Question: 23

Augmented matrix is given where asterisk is unspecified real number. Determine solution is consistent if so whether unique. No solution if not enough info.

a-

$$\left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{array}\right]$$

→ No of pivots = no of variables
→ No zero row
→ No contradiction.

↳ Consistent & Unique.

b-

$$\left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 0 \end{array}\right]$$

↳ Consistent & Inifinite

→ No contradiction
→ one free variable

c-

$$\left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 0 & 1 \end{array}\right]$$

↳ Inconsistent

→ contradiction.

d-

$$\left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 0 & * & 0 \\ 0 & 0 & 1 & * \end{array}\right]$$

↳ Not enough info

→ Row 2 depend on value of *

Question : 24

Augmented matrix is given where asterisk is unspecified real numbers. Determine solution is consistent if so whether unique. No solution if not enough info.

a-

$$\left[\begin{array}{cccc} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & 1 \end{array}\right]$$

↳ consistent & unique.

→ No of pivots = no of variables
→ No zero row
→ No contradiction

b-

$$\begin{bmatrix} 1 & 0 & 0 & * \\ * & 1 & 0 & * \\ * & * & 1 & * \end{bmatrix}$$

↳ Not enough info.

c-

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & * & * & * \end{bmatrix}$$

↳ Not Consistent

→ In row 1 $x_1 = 0$
→ In row 2 $x_1 = 1$
→ A variable can't be two different constant at same time.

d-

$$\begin{bmatrix} 1 & * & * & * \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Question : 25-26

Determine value of a for which system has no sol. exactly one sol or infinitely many solu.

$$\begin{aligned} x + 2y - 3z &= 4 \\ 3x - y + 5z &= 2 \\ 4x + y + (a^2 - 14)z &= a + 2 \end{aligned}$$

Taking Augmented Matrix
$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & 5 & 2 \\ 4 & 1 & a^2-14 & a+2 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & -7 & 14 & -10 \\ 0 & -7 & a^2-2 & a-14 \end{array}\right] \begin{array}{l} R_2-3R_1 \\ R_3-4R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & -2 & 10/7 \\ 0 & -7 & a^2-2 & a-14 \end{array}\right] -\frac{1}{7}R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & -16/7 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & a^2-16 & a-4 \end{array}\right] \begin{array}{l} R_1-2R_2 \\ R_3+7R_1 \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 16/7 \\ 0 & 1 & -2 & 10/7 \\ 0 & 0 & 1 & \frac{1}{a+4} \end{array}\right] \frac{R3}{a^2-16}$$

Infinite: If $a=4$

No Sol: If $a=-4$

Unique: If $a \neq \pm 4$

26-

$$\begin{aligned} x+2y+z &= 2 \\ 2x-2y+3z &= 1 \\ x+2y-(a^2-3)z &= a \end{aligned}$$

Taking Augmented Matrix.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 2 & -2 & 3 & 1 \\ 1 & 2 & -(a^2-3) & a \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & -6 & 1 & -3 \\ 0 & 0 & -a^2+2 & a-2 \end{array}\right] \begin{array}{l} R_2-2R_1 \\ R_3-R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 1 & -1/6 & 1/2 \\ 0 & 0 & -a^2+2 & a-2 \end{array}\right] \frac{R2}{-6} \quad \left[\begin{array}{cccc} 1 & 2 & 1 & 2 \\ 0 & 1 & -1/6 & 1/2 \\ 0 & 0 & 1 & \frac{a-2}{-a^2+2} \end{array}\right] \frac{R3}{-a^2+2}$$

~~Inf~~ Unique: if $a^2 \neq 2$

NoSol: if $a^2 = 2$

Infinite: No such value exist

Question : 27-28

What condition if any must a, b and c satisfy for the linear system to be consistent?

27

$$x+3y-z=a$$
$$x+y+2z=b$$
$$2y-3z=c$$

Taking Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 3 & -1 & a \\ 1 & 1 & 2 & b \\ 0 & 2 & -3 & c \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 3 & -1 & a \\ 0 & -2 & 3 & b-a \\ 0 & 2 & -3 & c \end{array}\right] R_2-R_1$$

$$\left[\begin{array}{ccc|c} 1 & 3 & -1 & a \\ 0 & 1 & -3/2 & a-b/2 \\ 0 & 2 & -3 & c \end{array}\right] \div \frac{R2}{2} \quad \left[\begin{array}{ccc|c} 1 & 3 & -1 & a \\ 0 & 1 & -3/2 & a-b/2 \\ 0 & 0 & 0 & c-(a-b) \end{array}\right] R_3-2R_2$$

For consistent

$$c-(a-b)=0$$
$$c-a+b=0$$

28

$$x+3y+z=a$$
$$-x-2y+z=b$$
$$3x+7y-z=c$$

Taking Aug matrix.

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ -1 & -2 & 1 & b \\ 3 & 7 & -1 & c \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & b+a \\ 0 & -2 & -4 & c-3a \end{array}\right] \begin{array}{l} R_2+R_1 \\ R_3-3R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & a \\ 0 & 1 & 2 & a+b \\ 0 & 0 & 0 & c-a+2b \end{array}\right] R_3+2R_2$$

For a System to be consistent
$c-a+2b = 0$

Question: 32
Reduce to reduce REF without introducing fraction at any intermediate step

$$\left[\begin{array}{cc|c} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 3 & 4 & 5 \end{array}\right]$$

$$\left[\begin{array}{cc|c} 2 & 1 & 3 \\ 0 & -2 & -29 \\ 0 & 5 & 1 \end{array}\right] 2R_3-3R_1 \quad \left[\begin{array}{cc|c} 2 & 1 & 3 \\ 0 & 2 & 29 \\ 0 & 5 & 1 \end{array}\right] -R_2 \quad \left[\begin{array}{ccc} 4 & 0 & -26 \\ 0 & 2 & 29 \\ 0 & 0 & -143 \end{array}\right] \begin{array}{l} 2R_3-5R_2 \\ 2R_1-R_2 \end{array}$$

$$\left[\begin{array}{ccc} 4 & 0 & -26 \\ 0 & 2 & 29 \\ 0 & 0 & 1 \end{array}\right] \frac{-R_3}{143} \quad \left[\begin{array}{ccc} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{array}\right] \begin{array}{l} R_2-29R_3 \\ R_1+26R_3 \end{array}$$

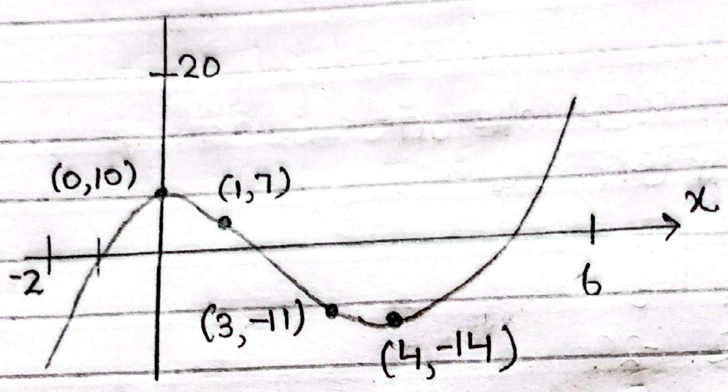
$$\left[\begin{array}{ccc} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \frac{R_2}{2} \quad \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \frac{R_1}{4}$$

↳ Identity Matrix.

Question : 37

Find cofficients a, b, c, d so that curve shown in Figure is graph of equation

$$y = ax^3 + bx^2 + cx + d \rightarrow (1)$$



Using point (0,10)

$10 = a(0)^3 + b(0)^2 + c(0) + d$

$d = 10$

Using (1,7)

$7 = a(1)^3 + b(1)^2 + c(1) + 10$

$7 = a + b + c + 10$

$a + b + c = -3$

Using (3,-11)

$-11 = a(3)^3 + b(3)^2 + c(3) + 10$

$-11 = 27a + 9b + 3c + 10$

$27a + 9b + 3c = -21$

Using (4,-14)

$-14 = a(4)^3 + b(4)^2 + c(4) + 10$

$-14 = 64a + 16b + 4c + 10$

$64a + 16b + 4c = -24$

So equations are.

$$a + b + c = -3$$
$$27a + 9b + 3c = -21$$
$$64a + 16b + 4c = -24$$

Taking Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -3 \\ 27 & 9 & 3 & -21 \\ 64 & 16 & 4 & -24 \end{array}\right] \quad \left[\begin{array}{ccc|c} 1 & 1 & 1 & -3 \\ 0 & -18 & -24 & 60 \\ 0 & -48 & -60 & 168 \end{array}\right] \begin{array}{l} R_2 - 27R_1 \\ R_3 - 64R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & -3 \\ 0 & 1 & 4/3 & -10/3 \\ 0 & -48 & -60 & 168 \end{array}\right] \; -\frac{R_2}{18} \qquad \left[\begin{array}{ccc|c} 1 & 0 & -1/3 & 1/3 \\ 0 & 1 & 4/3 & -10/3 \\ 0 & 0 & 4 & 8 \end{array}\right] \; \begin{array}{l} R_1 - R_2 \\ R_3 + 48R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1/3 & 1/3 \\ 0 & 1 & 4/3 & -10/3 \\ 0 & 0 & 1 & 2 \end{array}\right] \; \frac{R_3}{4} \qquad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -6 \\ 0 & 0 & 1 & 2 \end{array}\right] \; \begin{array}{l} R_2 - \frac{4}{3}R_3 \\ R_1 + \frac{1}{3}R_3 \end{array}$$

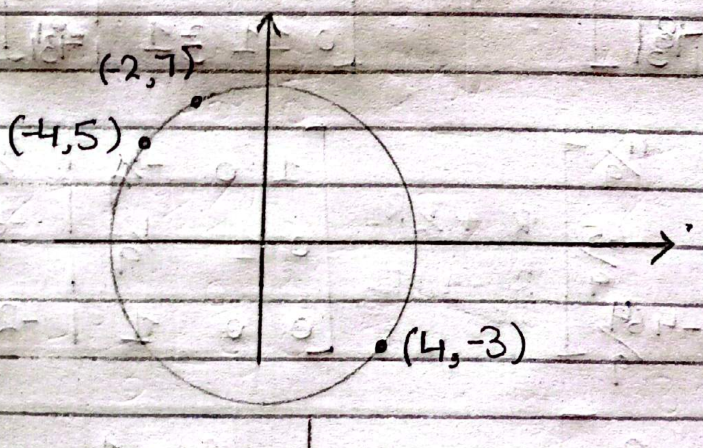
$\boxed{a = 1}$ $\boxed{b = -6}$ $\boxed{y = x^3 - 6x^2 + 2x + 10}$

$\boxed{c = 2}$ $\boxed{d = 10}$

Question : 38-

Find cofficient of a, b, c and d so that circle is given by equation.

$$ax^2 + ay^2 + bx + cy + d = 0$$



Using (-4,5)

$a(-4)^2 + a(5)^2 + b(-4) + c(5) + d = 0$

$16a + 25a - 4b + 5c + d = 0$

$41a - 4b + 5c + d = 0$

Using (-2,7)

$a(-2)^2 + a(7)^2 + b(-2) + c(7) + d = 0$

$4a + 49a - 2b + 7c + d = 0$

$53a - 2b + 7c + d = 0$

set $a = 1$ only to remove extra constant circle exactly remain same.

Using $(4,-3)$

$$a(4)^2+a(-3)^2+b(4)+c(-3)+d=0$$
$$16a+9a+4b-3c+d=0$$
$$25a+4b-3c+d=0$$

If we set $a=1$ so all eqs Become.

$$-2b+7c+d=-53$$
$$-4b+5c+d=-41$$
$$4b-3c+d=-25$$

Taking Augmented Matrix

$$\left[\begin{array}{ccc|c} -2 & 7 & 1 & -53 \\ -4 & 5 & 1 & -41 \\ 4 & -3 & 1 & -25 \end{array}\right] \qquad \left[\begin{array}{ccc|c} 1 & -7/2 & -1/2 & 53/2 \\ -4 & 5 & 1 & -41 \\ 4 & -3 & 1 & -25 \end{array}\right] \; \frac{-R_1}{2}$$

$$\left[\begin{array}{ccc|c} 1 & -7/2 & -1/2 & 53/2 \\ 0 & -9 & -1 & 65 \\ 0 & 11 & 3 & -131 \end{array}\right] \begin{array}{l} R_2+4R_1 \\ R_3-4R_1 \end{array} \qquad \left[\begin{array}{ccc|c} 1 & -7/2 & -1/2 & 53/2 \\ 0 & 1 & +1/9 & -65/9 \\ 0 & 11 & 3 & -131 \end{array}\right] \; \frac{-R_2}{9}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1/9 & 11/9 \\ 0 & 1 & 1/9 & -65/9 \\ 0 & 0 & 16/9 & -464/9 \end{array}\right] \begin{array}{l} R_1+\frac{7}{2}R_2 \\ R_3-11R_2 \end{array} \qquad \left[\begin{array}{ccc|c} 1 & 0 & -1/9 & 11/9 \\ 0 & 1 & 1/9 & -65/9 \\ 0 & 0 & 1 & -29 \end{array}\right] \; \frac{9R_3}{16}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & -29 \end{array}\right] \begin{array}{l} R_2-\frac{1}{9}R_3 \\ R_1+\frac{1}{9}R_3 \end{array}$$

$\boxed{a=1}$

$\boxed{b=-2}$

$\boxed{c=-4}$

$\boxed{d=-29}$

$$\boxed{x^2+y^2-2x-4y-29=0}$$

Question no 4

(a) If A is a matrix with three rows and five columns, then what is the maximum possible number of leading 1's in its reduced row echelon form?

The matrix has 3 rows and 5 columns.

In Reduced Row Echelon Form (RREF):

- Each row can have only one leading 1.
- So, the number of leading 1's cannot be more than the number of rows.

Since there are 3 rows,

Maximum number of leading 1's = 3.

(b) If B is a matrix with three rows and six columns, then what is the maximum possible number of parameters in the general solution of the linear system with augmented matrix B?

Matrix B has:

- 3 rows
- 6 columns

Since it is an augmented matrix, the last column is constants.

So, number of variables = $6 - 1 = 5$

Maximum rank (maximum leading 1's) = 3 (because there are 3 rows)

Number of parameters =

$$\begin{aligned}&\textbf{Number of variables} - \textbf{Rank}\\&= \mathbf{5 - 3 = 2}\end{aligned}$$

(c) If C is a matrix with five rows and three columns, then what is the minimum possible number of rows of zeros in any row echelon form of C?

Matrix C has:

- 5 rows
- 3 columns

Maximum possible leading 1's (rank) = 3
(because rank cannot be more than the number of columns)

So, at most 3 rows can be non-zero.

Since total rows = 5

Minimum zero rows =

$$\mathbf{5 - 3 = 2}$$

True-False Exercises

a) If a matrix is in reduced row echelon form, then it is also in row echelon form.

True. A matrix in reduced row echelon form satisfies all conditions of row echelon form (and additional constraints).

b) If an elementary row operation is applied to a matrix that is in row echelon form, the resulting matrix will still be in row echelon form.

False. An elementary row operation can destroy row echelon properties (e.g., swapping rows can move a leading 1 improperly).

c) Every matrix has a unique row echelon form.

False. Every matrix has a unique reduced row echelon form; row echelon form is not unique.

d) A homogeneous linear system in n unknowns whose corresponding augmented matrix has a reduced row echelon form with r leading 1's has n − r free variables

True

Explanation

- n = total number of variables
- r = number of leading 1's (rank)
- Leading 1's represents basic variables
- Remaining variables are free variables (parameters)

So,

$$\text{Free variables} = n - r$$

Example

Suppose:

- n = 5 variables
- r = 3 leading 1's

Then:

$$5 - 3 = 2$$

So, there are 2 free variables

e) All leading 1's in a matrix in row echelon form must occur in different columns.

True. By definition, each leading 1 in row echelon form is in a distinct column.

f) If every column of a matrix in row echelon form has a leading 1, then all entries that are not leading 1's are zero.

False. In row echelon form (not reduced), entries above the leading 1's can be non-zero. Only in reduced row echelon form do all other entries become zero

g) If a homogeneous linear system of n equations in n unknowns has a corresponding augmented matrix with a reduced row echelon form containing n leading 1's, then the linear system has only the trivial solution.

True. A homogeneous system with n equations, n unknowns, and n leading 1's has only the trivial solution.

h) If the reduced row echelon form of the augmented matrix for a linear system has a row of zeros, then the system must have infinitely many solutions.

False. Row of zeros in RREF only means no contradiction. It does not guarantee infinitely many solutions.

i) If a linear system has more unknowns than equations, then it must have infinitely many solutions.

False. If a linear system has more unknowns than equations, it has either no solution or infinitely many (if consistent).